

Data and Artificial Intelligence

Cyber Shujaa Program

Week 3 Assignment Titanic Exploratory Data Analysis

Student Name: Hope Njoki Nyambura

Student ID: cs-eh03-25100

Table of Contents

1. Introduction.....	2
2. Objectives	2
3. Tasks Completed.....	2
Step 1: Import Libraries	2
Step 2: Load Dataset	3
Step 3: Initial Data Exploration	3
Step 4: Handling Missing Values and Outliers	4
Step 5: Univariate Analysis.....	4
Step 6: Bivariate Analysis	5
Step 7: Multivariate Analysis.....	5
Step 8: Target Variable Analysis	6
4. Conclusion	6
5. Link to Kaggle Notebook.....	6

1. Introduction

This assignment focused on performing Exploratory Data Analysis (EDA) using the Titanic dataset from Kaggle.

The purpose was to gain practical experience in profiling data, handling missing values, analyzing distributions, detecting outliers, and exploring feature relationships that influence survival on the Titanic.

2. Objectives

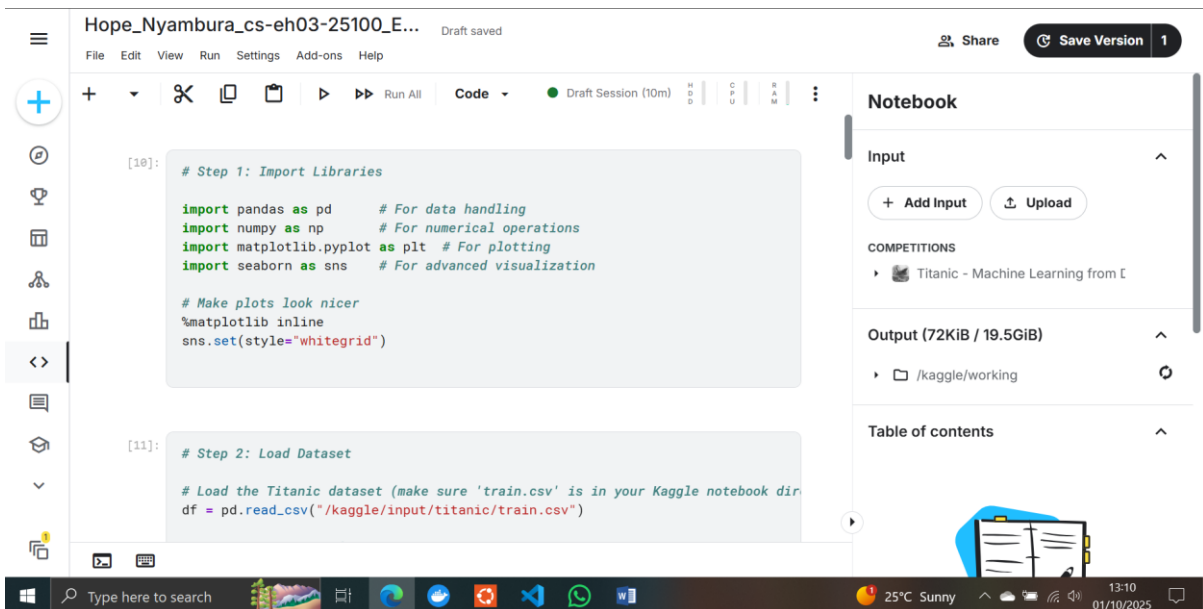
The main objectives of this assignment were:

- To explore the Titanic dataset and understand its structure.
- To assess and handle missing values, duplicates, and outliers.
- To conduct univariate, bivariate, and multivariate analysis.
- To analyze the survival variable and identify key factors influencing survival.
- To publish the final notebook on Kaggle.

3. Tasks Completed

Step 1: Import Libraries

- Imported Pandas, NumPy, Matplotlib, and Seaborn for analysis and visualization.



The screenshot displays a Jupyter Notebook interface. The top bar shows the notebook name 'Hope_Nyambura_cs-eh03-25100_E...' and a 'Draft saved' status. The left sidebar contains navigation icons. The main area shows two code cells. The first cell, labeled '[10]:', is titled '# Step 1: Import Libraries' and contains the following code:

```
import pandas as pd # For data handling
import numpy as np # For numerical operations
import matplotlib.pyplot as plt # For plotting
import seaborn as sns # For advanced visualization

# Make plots look nicer
%matplotlib inline
sns.set(style='whitegrid')
```

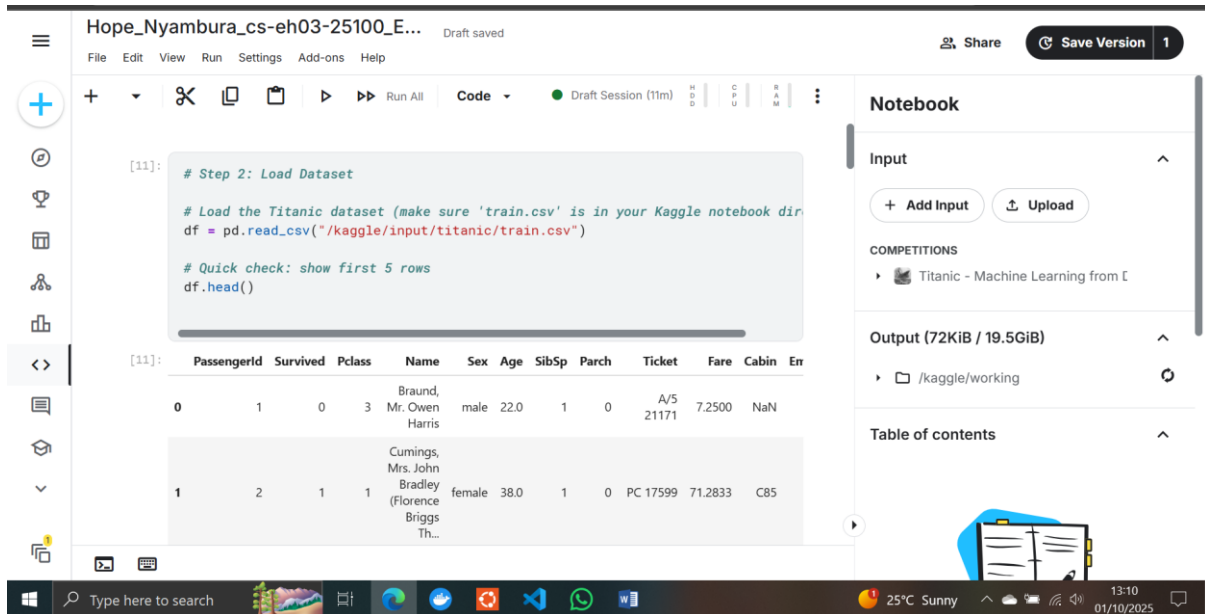
. The second cell, labeled '[11]:', is titled '# Step 2: Load Dataset' and contains the code:

```
# Load the Titanic dataset (make sure 'train.csv' is in your Kaggle notebook dir)
df = pd.read_csv("/kaggle/input/titanic/train.csv")
```

. The right sidebar shows the 'Notebook' panel with 'Input' (Add Input, Upload), 'COMPETITIONS' (Titanic - Machine Learning from E), 'Output (72KiB / 19.5GiB)' (/kaggle/working), and 'Table of contents'.

Step 2: Load Dataset

- Loaded the Titanic dataset into a Pandas DataFrame.
- Displayed the first few rows using `df.head()`.



```
[11]: # Step 2: Load Dataset

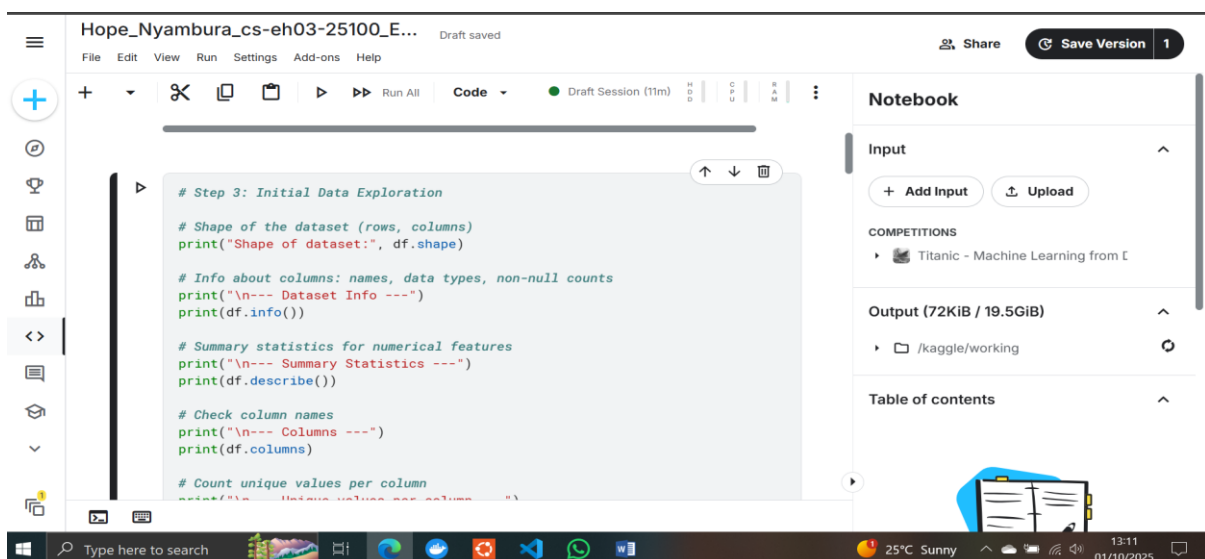
# Load the Titanic dataset (make sure 'train.csv' is in your Kaggle notebook dir)
df = pd.read_csv("/kaggle/input/titanic/train.csv")

# Quick check: show first 5 rows
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.2833	C85	

Step 3: Initial Data Exploration

- Used:
 - `df.shape` → to get rows and columns.
 - `df.info()` → to check data types and missing values.
 - `df.describe()` → to summarize numerical features.
 - `df.duplicated().sum()` → checked duplicate rows.
 - `df.nunique()` → counted unique values per column.
- Documented feature list (PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked).



```
# Step 3: Initial Data Exploration

# Shape of the dataset (rows, columns)
print("Shape of dataset:", df.shape)

# Info about columns: names, data types, non-null counts
print("\n--- Dataset Info ---")
print(df.info())

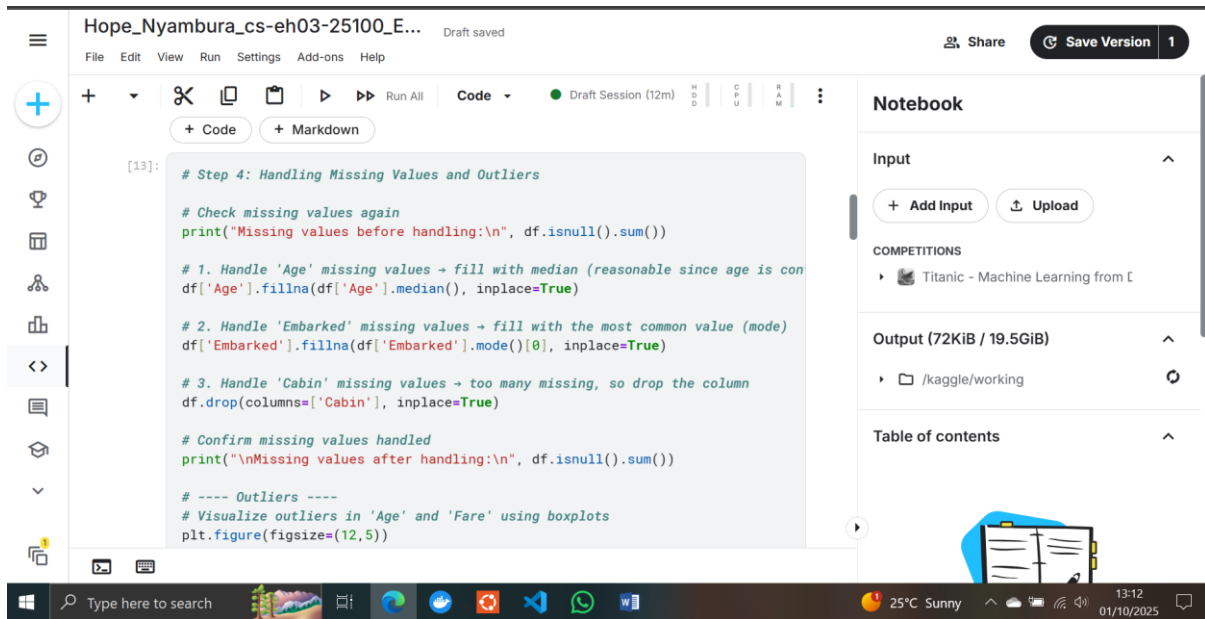
# Summary statistics for numerical features
print("\n--- Summary Statistics ---")
print(df.describe())

# Check column names
print("\n--- Columns ---")
print(df.columns)

# Count unique values per column
print(df.nunique())
```

Step 4: Handling Missing Values and Outliers

- Missing values:
 - Age → imputed with median.
 - Cabin → dropped due to high missingness.
 - Embarked → filled with most common port.
- Outliers:
 - Detected using boxplots for Age and Fare.
 - Decided to keep extreme fares (important for wealth insights).



```
[13]: # Step 4: Handling Missing Values and Outliers

# Check missing values again
print("Missing values before handling:\n", df.isnull().sum())

# 1. Handle 'Age' missing values → fill with median (reasonable since age is con
df['Age'].fillna(df['Age'].median(), inplace=True)

# 2. Handle 'Embarked' missing values → fill with the most common value (mode)
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)

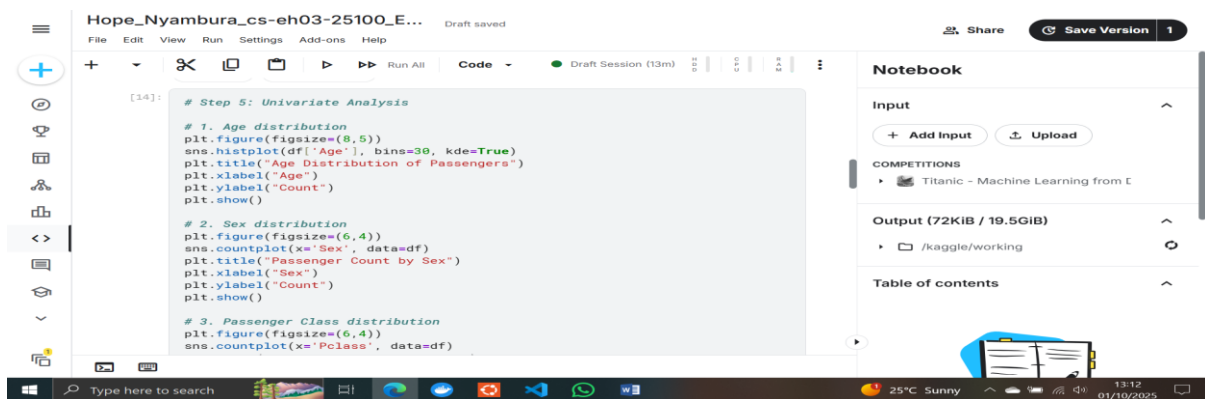
# 3. Handle 'Cabin' missing values → too many missing, so drop the column
df.drop(columns=['Cabin'], inplace=True)

# Confirm missing values handled
print("\nMissing values after handling:\n", df.isnull().sum())

# ---- Outliers ----
# Visualize outliers in 'Age' and 'Fare' using boxplots
plt.figure(figsize=(12,5))
```

Step 5: Univariate Analysis

- Age distribution → histogram.
- Sex counts → bar plot.
- Pclass distribution → count plot.
- Fare distribution → histogram (skewed).
- Embarked distribution → count plot.



```
[14]: # Step 5: Univariate Analysis

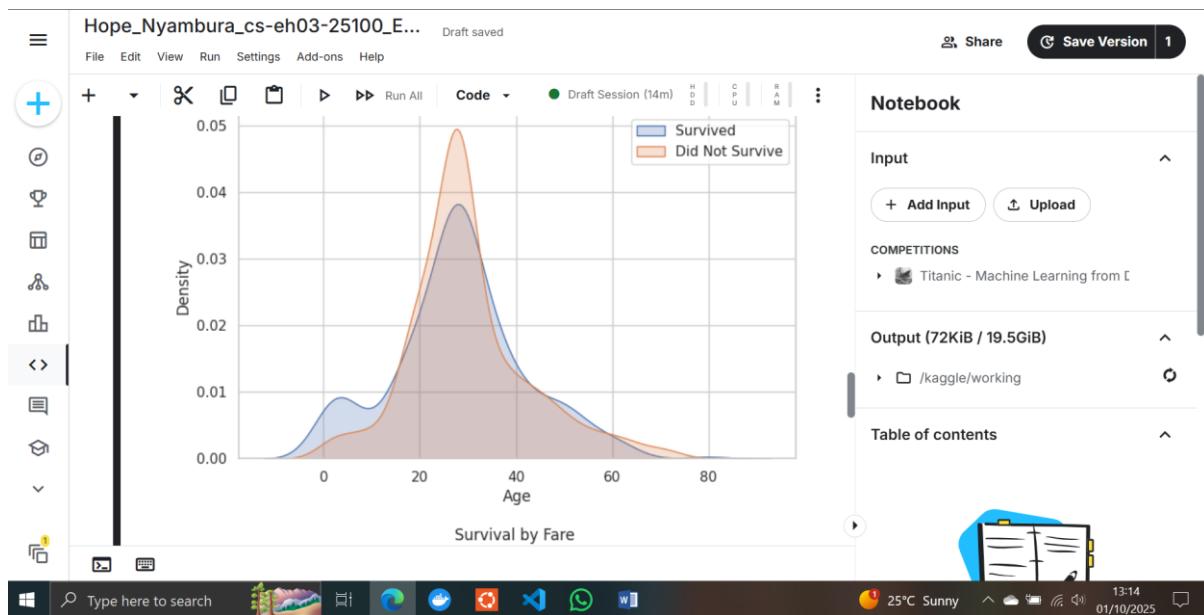
# 1. Age distribution
plt.figure(figsize=(8,5))
sns.histplot(df['Age'], bins=30, kde=True)
plt.title("Age Distribution of Passengers")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()

# 2. Sex distribution
plt.figure(figsize=(6,4))
sns.countplot(x='Sex', data=df)
plt.title("Passenger Count by Sex")
plt.xlabel("Sex")
plt.ylabel("Count")
plt.show()

# 3. Passenger Class distribution
plt.figure(figsize=(6,4))
sns.countplot(x='Pclass', data=df)
```

Step 6: Bivariate Analysis

- Survival vs Sex → count plot.
- Survival vs Pclass → bar plot.
- Survival vs Age → KDE plot.
- Survival vs Embarked → bar plot.
- Survival vs Fare → violin plot.



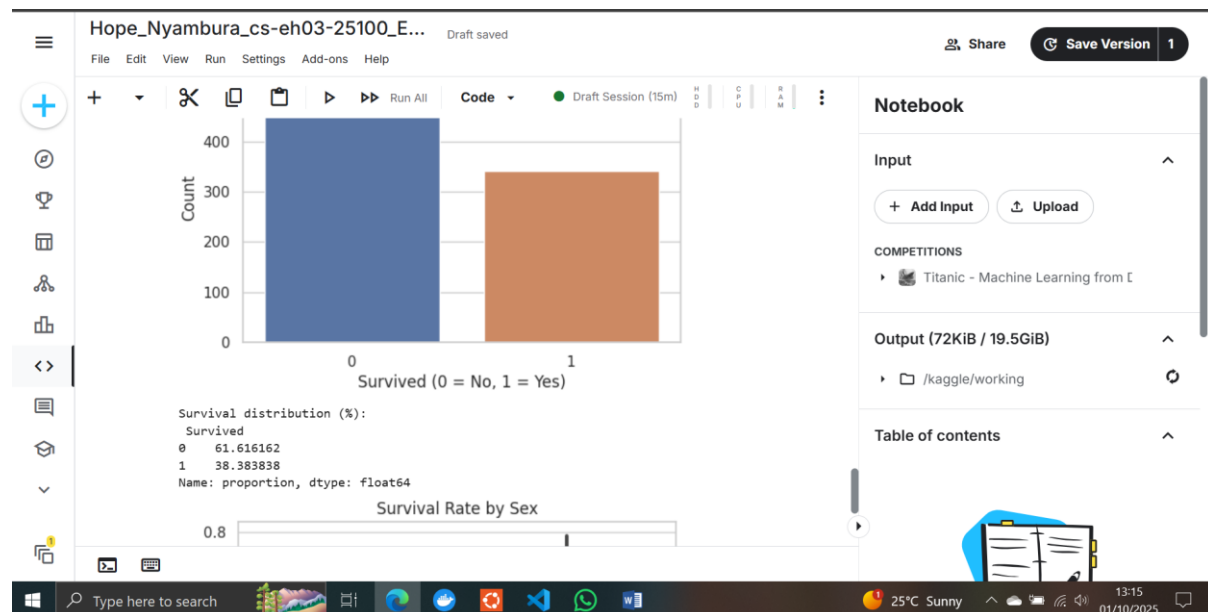
Step 7: Multivariate Analysis

- Pclass, Sex, and Survival → grouped bar plot.
- Pclass, Age, Fare, and Survival → scatter plots.
- Embarked, Pclass, and Survival → stacked plots.



Step 8: Target Variable Analysis

- Target = Survived (0 = No, 1 = Yes).
- Dataset is imbalanced (~62% did not survive, ~38% survived).
- Key findings:
 - Women had higher survival rates.
 - Higher class passengers had higher survival.
 - Younger children had higher chances of survival.



4. Conclusion

This assignment provided valuable practice in exploratory data analysis using Pandas, Matplotlib, and Seaborn.

Key insights include:

- Gender and class were strong predictors of survival.
 - Fare and age showed important but less direct relationships.
 - Embarkation point also influenced survival, particularly when combined with class.
- The dataset is now well-understood and ready for further predictive modeling.

5. Link to Kaggle Notebook

[<https://www.kaggle.com/code/defhopeee/hope-nyambura-cs-eh03-25100-eda>]